



Rapidly Deployable, Mission-Critical Modular Infrastructure

*Engineered with Patented Northstar Composite Technology
Servicing Enterprise, Defense and Infrastructure Markets*



PRODUCT OVERVIEW

*Purpose-built modular infrastructure for compute, power, and security
— engineered to deploy where traditional construction cannot*

The Synapse Module is a factory-integrated, mission-critical, modular infrastructure platform engineered by Northstar Building Systems. Each unit ships as a complete, turnkey system; structural enclosure, power distribution, thermal management, and network-ready architecture. 100% US-Made & manufactured under controlled factory conditions, delivered within 120 days of design approval and operational within 2 weeks of site delivery.

The structural shell is built on Northstar's Advanced Fiber Reinforced Polymer (FRP) composite system (Patented), an aero-space grade material that is pound-for-pound stronger than steel at 75% less weight, non-conductive, non-magnetic, non-corrosive, and fire-rated to ASTM E119. Our technology's advanced properties are what make Synapse natively compatible with TEMPEST, SCIF, and RF-sensitive environments, without retrofits or add-on shielding.

7 purpose-built series address distinct mission profiles; from standard enterprise IT (8–12 kW/rack) to next-gen AI supercomputing (up to 200 kW/rack with direct liquid cooling). Platform configurations span 8' to 12' wide and 10' to 50' lengths, available in non-ISO and ISO-standard form factors for logistics flexibility.

Why FRP Changes the Infrastructure Equation:

250 MPH

Wind Rating

50+ YR

Design Life

14 WK

Deploy Time

38%

HVAC Savings



TECHNICAL SPECIFICATIONS

PARAMETER	SPECIFICATION
Platform Configs	12' Wide 8' Wide Non-ISO ISO Standard ISO High Cube
Model Series	G (Enterprise) H (HPC) AI (Training) F (DoD) XL (Next-Gen AI) T (Telecom) E (Edge)
12' Wide Lengths	10' 20' 30' 40' 50' (all 12' W x 12' H)
8' Wide Lengths	10' 20' 40' (all 8' W x 8' H)
Structural	350lbs psf floor load 250 mph wind rating IBC roof ratings
Power	208V / 480V 3-phase 100A–800A N+1 redundancy options
Rack Density	8–12 kW (G) 20–40 kW (H) 50–80 kW (AI) Up to 200 kW (XL)
Cooling	Standard DX/CRAH In-Row/RDHx Liquid-to-Chip CDU Direct Liquid + Rear Door
Temp. Range	–40 °F to +140 °F (–40 °C to +60 °C)
IP / Fire	IP55 std (IP66 Available) ASTM E-84 Class A (E-119 Available) UL Class A roof
Blast / Ballistic	HVHZ Missile Impact: Level D UL 752 Standard Level 1 (2 - 8 Available)
Certifications	UL NIST 800-171 (CUI Ready) DoD STIG TEMPEST/SCIF ICD 705 Intertek
Deployment	4–8 months design-to-deploy 7–21 days delivery-to-operational

All specifications configurable per mission requirements. Contact NED for custom configurations.

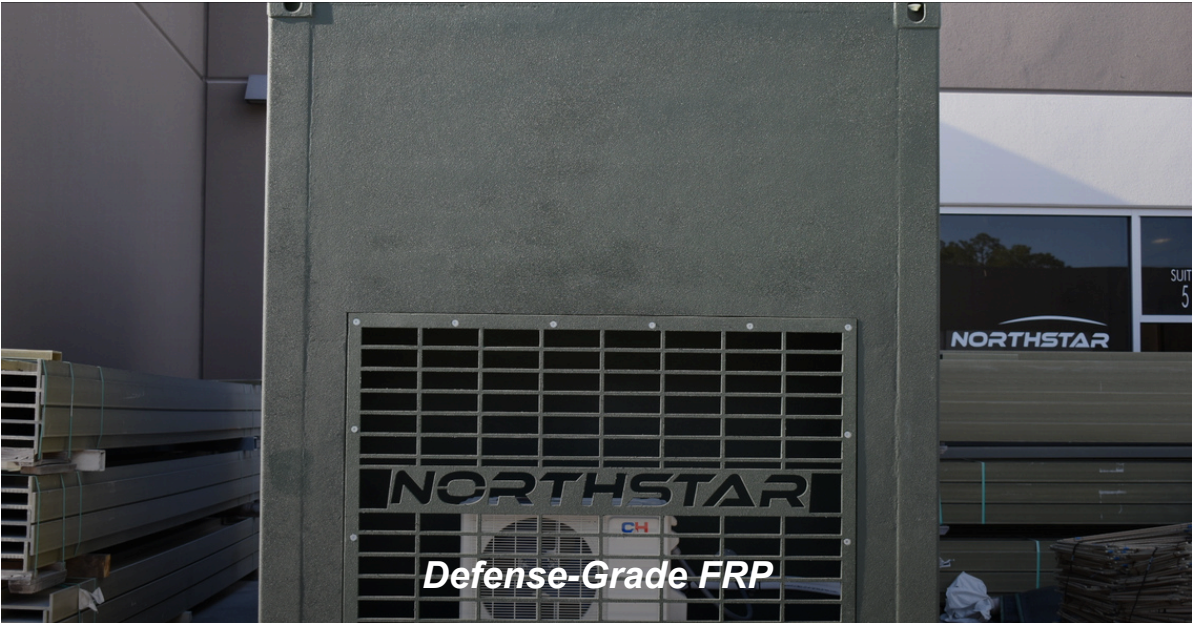
SYNAPSE MODEL TYPES

Seven specialized series engineered for distinct mission profiles across enterprise, defense, and government sectors.

SERIES	CODE	USE CASE	kW/RACK	COOLING
General Enterprise	G	Standard IT / colocation	8–12 kW	Standard DX / CRAH
High Performance	H	HPC / high-density compute	20–40 kW	In-Row / Rear Door HX
AI Training	AI	AI/ML training clusters	50–80 kW	Liquid-to-Chip CDU
Federal / DoD	F	Classified / SCIF / tactical	10–15 kW	Ruggedized DX / CRAH
Next-Gen AI	XL	AI supercomputing	Up to 200 kW	Direct Liquid + Rear Door
Telecom	T	5G / ILA / macro cell	5–8 kW	Ruggedized DX
Edge Inference	E	Edge AI inference	30–60 kW	In-Row / Liquid CDU

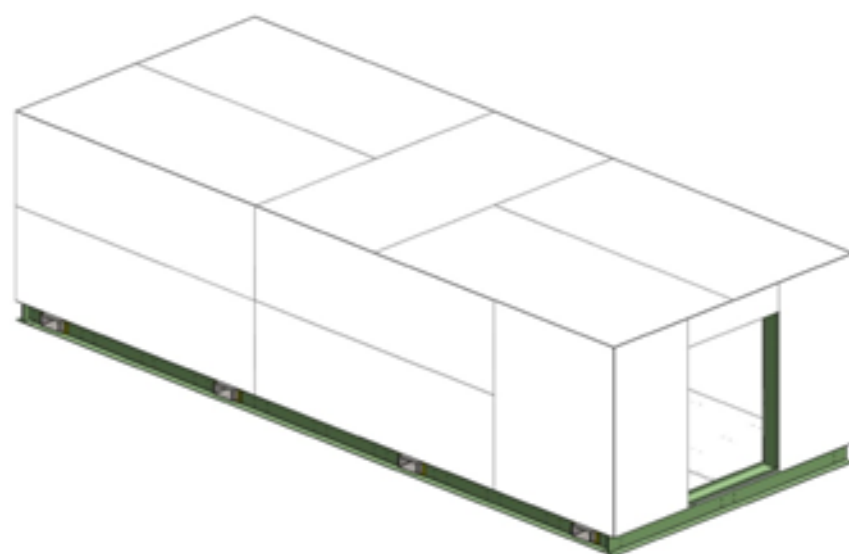
PLATFORM CONFIGURATIONS

12' Wide: 10' | 20' | 30' | 40' | 50' (12"W x 12'H)
8' Wide Non-ISO: 10' | 20' | 40' (8"W x 8'H)
ISO Standard: 20' | 40' (8"W x 8'6"H)
ISO High Cube: 20' | 40' (8"W x 9'6"H)

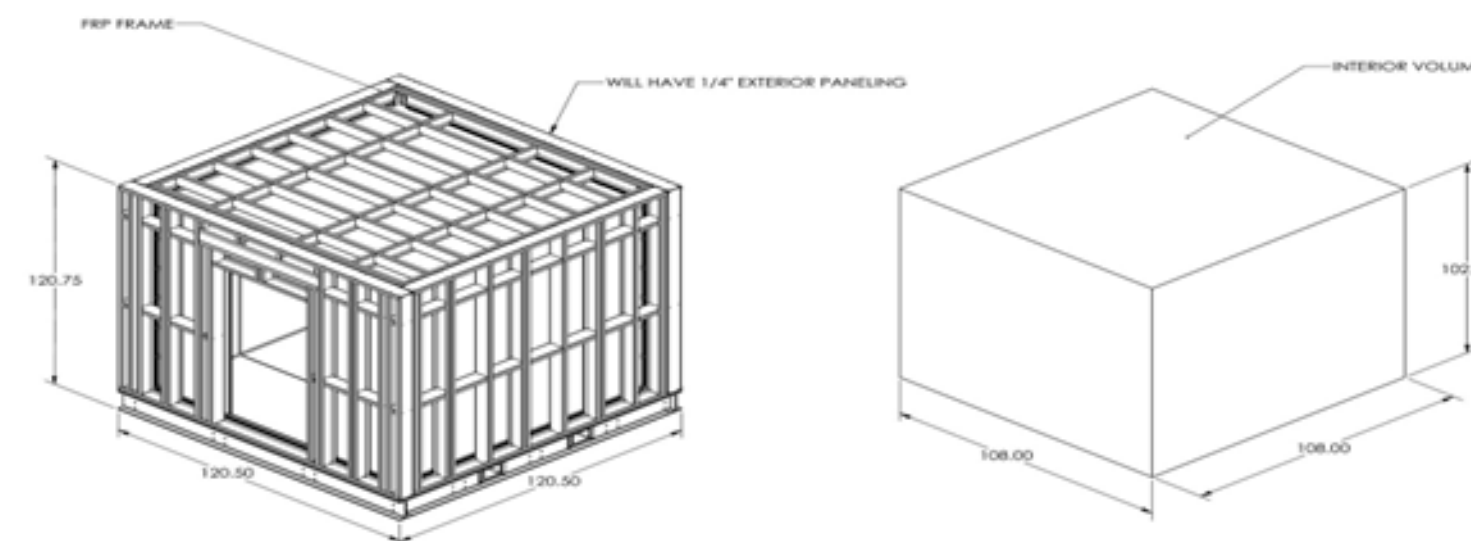


STRUCTURAL ENGINEERING MODELS

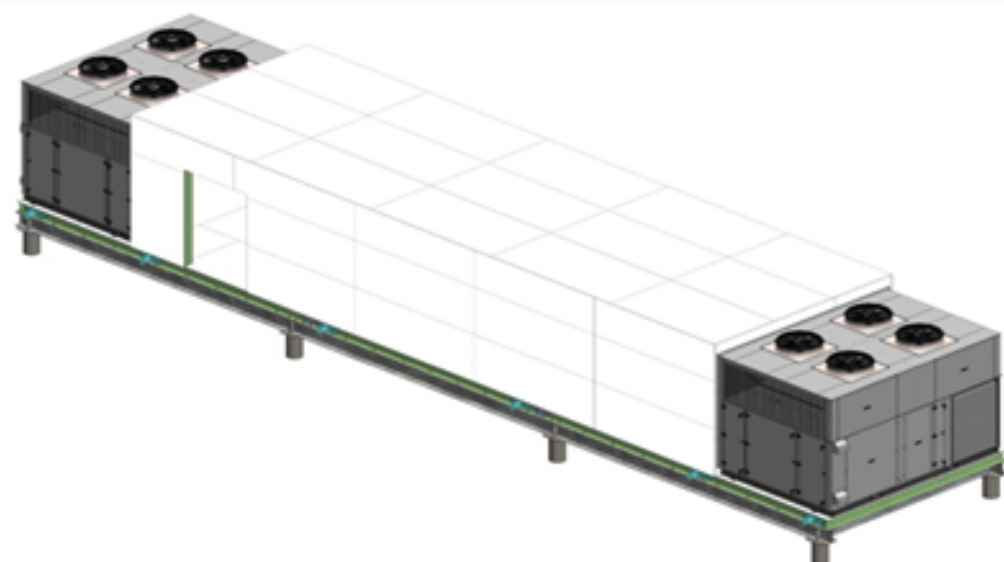
NAMING CONVENTION: Synapse [Length]-[Series Code] — Example: Synapse 30-AI = 30' AI Training module | Synapse 820-T = 8' x 20' Telecom module



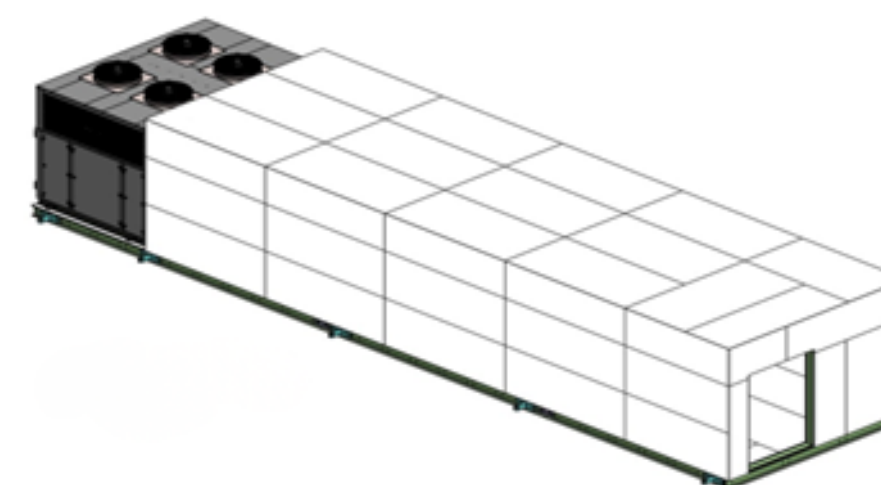
Synapse 820-T — Isometric Assembly



Synapse 10-G — FRP Frame + Interior Volume



Synapse 55-AI — Plan View



Synapse 40-H — Plan View

Drawings are representative. Final engineering documentation provided with each order. All structures reference PFRP Wall Panel Assembly per U.S. Patent No. 11,613,891.

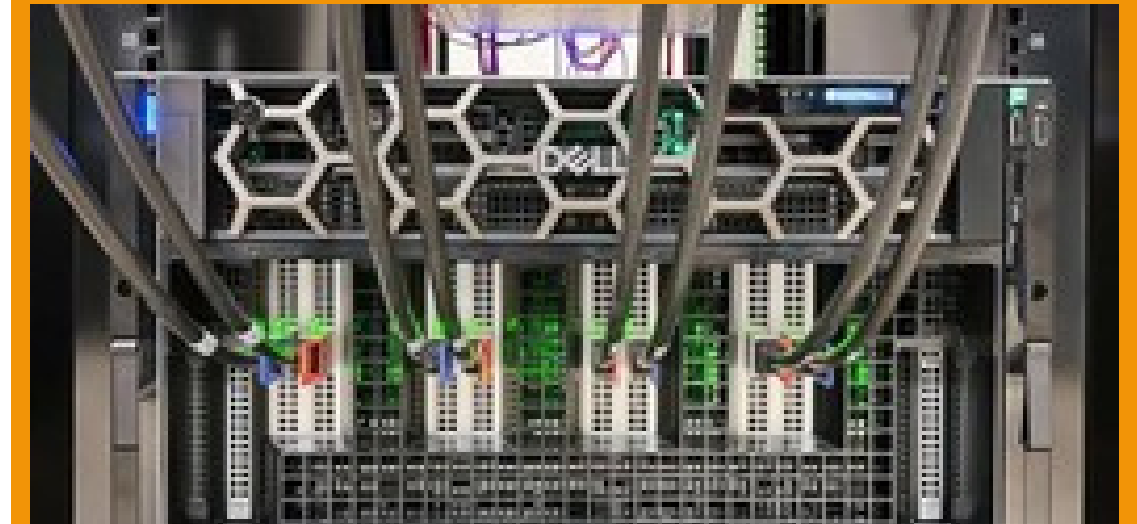
SYNAPSE PRODUCT GALLERY



Enterprise Synapse Module



Defense-Grade Synapse Module



High-Density Server Rack Interior



Mobile Synapse Deployment



Crane Installation



FRP Module Interior

Additional photo and video galleries available on our websites at the end of this document

NORTHSTAR FIBER REINFORCED POLYMER TECHNOLOGY

Northstar Technologies Group has developed a patented fiber reinforced polymer (FRP) composite building system that redefines the future of construction. Engineered for strength, speed, and sustainability, our advanced system utilizes precision-manufactured FRP components to create complete building structures—from the foundation to the roof.

Features:



Fire Resistant

ASTM E84 Class A fire rating & ASTM E119 1-hour resistance



Ballistic Rated

UL 752 ballistic tested up to Level 8 (.50 cal)



Corrosion Resistant

Zero water/air infiltration, non-porous, termite-resistant



Impact Rated

250+ mph impact rating (HVHZ certified, hurricane resilience)



Light-weight, Yet High-Strength

FRP is pound-for-pound stronger than steel in lengthwise direction, while weighing 75% less than steel. Its strength-to-weight ratio is 4X that of high tensile steel.

EXOSHELL™ Wall System

8th-generation composite ballistic armor plate bonded to a pultruded FRP frame assembly. Factory-fitted, sealed, and fire-coated. Zero water penetration. Zero air permeability



NFPA 285 Fire Test



Northstar's FRP during assembly



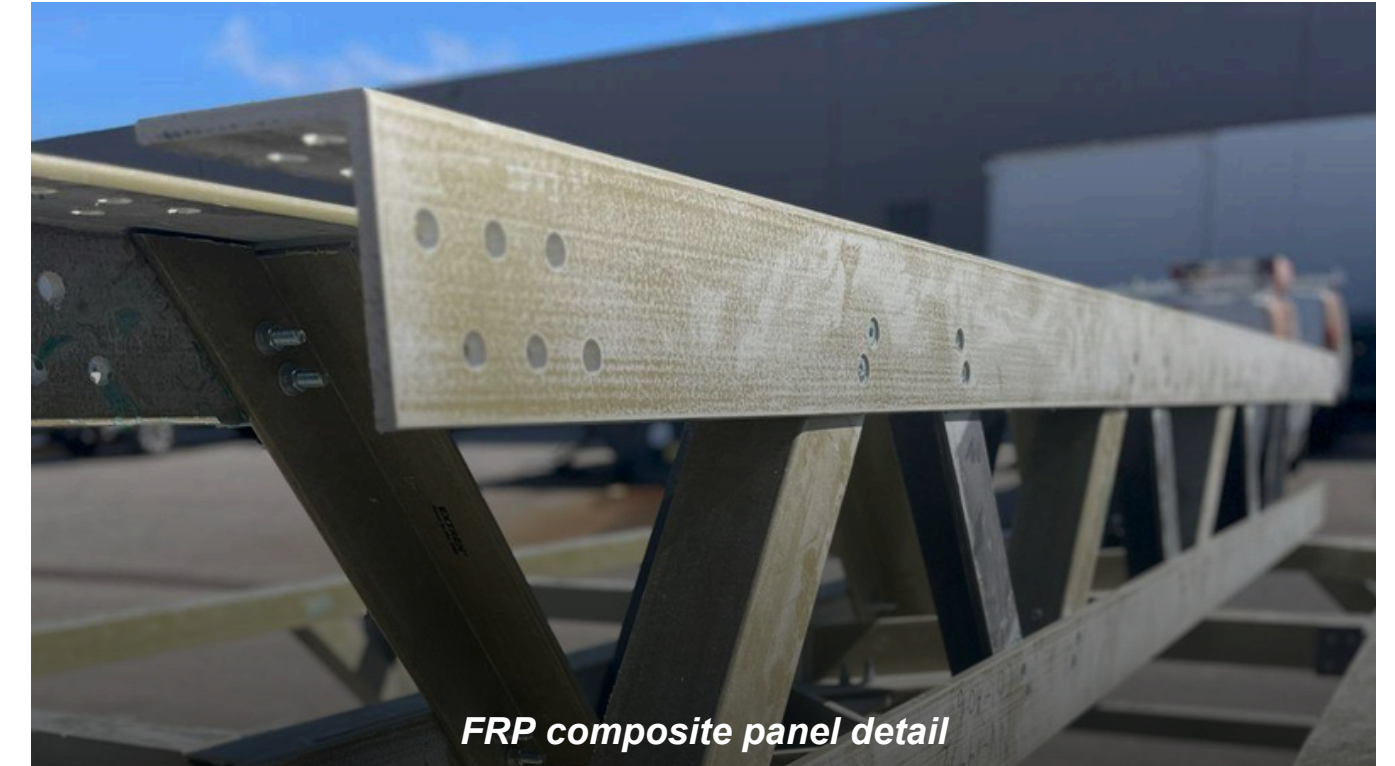
Ballistic Panel Stopping 9mm Rounds

MODULAR COMPOSITE ADVANTAGE

We engineered Synapse using advanced Fiber Reinforced Polymer (FRP) materials to deliver stronger, lighter, and more resilient infrastructure for modern data center demands. FRP enables rapid deployment, superior durability, and resistance to corrosion, fire, and extreme environmental conditions, ensuring reliable modular compute, power, and network solutions that move at the speed of your mission.

Whether enabling AI at the edge, scaling LLM and SLM workloads, protecting critical systems, or deploying full digital infrastructure in days instead of months, Northstar's FRP provides the flexibility, longevity, and performance today's operations require.

- ✓ **Corrosion Resistance**
No rust, no rot. Non-corrosive, moisture-resistant, pest-proof. Ideal for coastal, humid, and chemical environments.
- ✓ **Non-Conductive & Non-Magnetic**
RF-transparent walls support 5G/IoT. Critical for SCIF, TEMPEST, and sensitive electronics.
- ✓ **80%+ Lighter Than Steel/Concrete**
Dramatically reduces transport costs, foundation requirements, and crane complexity.
- ✓ **Superior Strength-to-Weight**
Exceeds steel pound-for-pound. 250 mph wind rating. Fire-resistant per ASTM E119.
- ✓ **50+ Year Design Lifespan**
Minimal maintenance. Supports disassembly and reuse — circular economy principles.
- ✓ **Rapid Manufacturing**
CNC-fabricated with integrated MEP raceways. Factory-controlled quality eliminates field variability.



FRP composite panel detail



Defense-grade FRP module exterior

COOLING & POWER INFRASTRUCTURE

Turnkey thermal management and power delivery — factory-integrated, mission-ready, and scalable to any density.

COOLING SOLUTIONS

Standard DX / CRAH

G-Series | F-Series

Direct expansion and Computer Room Air Handler units. Precision air cooling for standard-density IT loads (8–15 kW/rack). Factory-ducted hot/cold aisle containment.

In-Row / Rear Door Heat Exchanger (RDHx)

H-Series | E-Series

Rack-adjacent cooling for high-density compute (20–60 kW/rack). Chilled water or DX variants. Active RDHx units tested to 70+ kW per cabinet.

Liquid-to-Chip CDU (Coolant Distribution Unit)

AI-Series

Direct-to-chip cold plates deliver coolant to GPUs and CPUs for AI/ML training clusters (50–80 kW/rack). Separate secondary loop via CDU isolates IT from facility water.

Direct Liquid + Rear Door (Hybrid)

XL-Series

Combined direct liquid cooling with rear door heat exchangers for next-gen AI supercomputing at up to 200 kW/rack. Pumped refrigerant economization available.

Ruggedized DX

T-Series | F-Series

Hardened cooling for telecom ILAs, remote sites, and tactical environments. Operates –40°F to +140°F. IP55/IP66 rated enclosures.

POWER INFRASTRUCTURE

Power Distribution

208V / 480V 3-phase power delivery. 100A to 800A service capacity per module. Factory-integrated switchgear, PDUs, and busway distribution. N+1 redundancy options standard on all configurations.

Transformer Solutions

Pad-mounted and skid-mounted transformer packages from 500 kVA to 10 MVA+. Medium-voltage (12.47 kV / 34.5 kV) to low-voltage step-down. Integrated with automatic transfer switches (ATS) for seamless utility and generator failover.

Microgrid — NED Supplied

NED provides turnkey microgrid solutions — self-contained power generation and management for off-grid, austere, or redundancy-critical deployments. Diesel, natural gas, and hybrid generator sets. Battery Energy Storage Systems (BESS) for peak shaving and UPS backup. Solar-ready integration for renewable hybrid configurations. Intelligent load management and automated switchover.

UPS & Backup Power

Online double-conversion UPS systems sized per module. Lithium-ion or VRLA battery strings. Scalable runtime from minutes to hours. Generator-ready interfaces with automatic transfer and paralleling.

208V/480V
3-Phase Power

800A
Max Service

10 MVA+
Transformer Cap.

N+1
Redundancy

38% HVAC REDUCTION vs. steel containers — FRP composite eliminates thermal bridging and reduces cooling load.

APPLICATIONS

Modular Data Centers & Edge Computing

Deploy high-density compute at the edge with full power, cooling, connectivity. Configs from 8' to 50'.

SCIF / Secure Compartmented Facilities

ICD 705 and TEMPEST compliant. Non-conductive FRP + composite RF shielding. F-series for DoD.

Tactical & Deployable Command Centers

Relocatable C4ISR capability. Signature-managed, dispersed nodes. Fight through DDIL.

Disaster Recovery & Emergency

Telecom / 5G Infrastructure

T-series for ILAs, remote sites, macro/small cell, CBRS. Ruggedized DX cooling.

Military FOB Infrastructure

Replace tent-based and ISO containers. Standard interfaces for rapid swap and relocation.

AI / HPC & Supercomputing

AI-series: 50–80 kW/rack, liquid CDU. XL-series: up to 200 kW/rack, direct liquid cooling.



Deployed in under 90 days. Built to last 50+ years.

Factory-to-operational in weeks. HVAC loads cut 38% vs. steel. 250 mph wind rated. Patented FRP technology. Turnkey leased solutions available to reduce CapEx.

PROVEN CASE STUDIES

Fortune 500 Telecom Company

Emergency Comms Center (Synapse T-40)

Rapidly deployable emergency communications center. Plug-and-play network integration. 250 mph wind rated. Deployed in 15 weeks.

Publicly Traded Bitcoin Miner

AI Modular Data Center (Synapse AI-40)

Delivered in 14 weeks. CNC-fabricated FRP shells cut HVAC loads by 38%. Flexes between mining and AI workloads.

Fortune 100 Energy Company

Enterprise Modular Data Center (Synapse E-50)

Redundant power, composite floors with 350lbs psf loading capacity, RF-transparent walls for 5G/IoT Data center capability where traditional construction takes 18–24 months. Delivered in 90 Days

United States Marine Corps

Tactical Deployment (Synapse ISO F-20)

Hardened mobile command unit. HVHZ Missile Impact Level D, UL 752 Standard Level 1 ballistic rating. TEMPEST/SCIF-Ready. Transported by standard military vehicles. Delivered in 30 days.

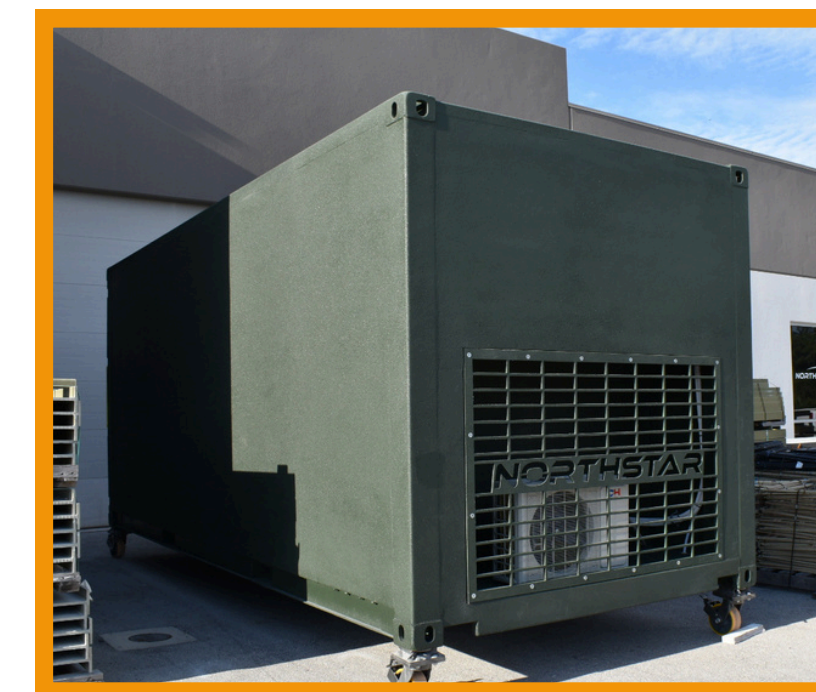
Department of Defense

Southern Command Center (Modular Equipment Housing)

350 mph impact resistance. Cat 5+ hurricane rated. Hybrid FRP modular equipment housing for corrosion-proof performance.



Telecom module deployment



Defense-Rated deployment



Southern Command Center, Doral FL

Additional Case Studies listed on www.northstarenterprisedefense.com/projects

READY TO BUILD THE FUTURE?

“Turnkey digital infrastructure — compute + network + security — anywhere. In Months, not years”

- Patented FRP Technology — lighter, stronger, longer-lasting than steel
- 14-Week Design-to-Deploy Track Record
- 38% HVAC Reduction vs. Steel Containers
- 250 MPH Wind Rating — Category 5+ Hurricane Resilience
- Mission-Ready — Factory-Integrated Power, Cooling, Connectivity
- 7 Specialized Series for Every Mission Profile

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FRP Building System: U.S. Patent No. 11,613,891.

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LET'S TALK

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